# State Diagram - Real-Time Audio Spectrum Analyzer (STM32F401)

After start-up the firmware sits in one ACTIVE state; six independent regions change its behaviour

**RESET**
power on / ST-Link reset

→

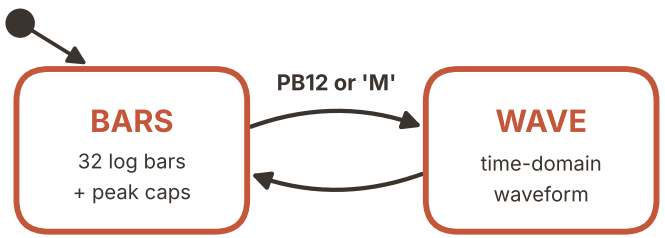
**INITIALISE**
clock 84 MHz > SysTick > GPIO > buttons > ADC+TIM2 > TFT > Bluetooth

→

**ACTIVE (main loop)**
sampling runs in ADC interrupt, work runs in main loop

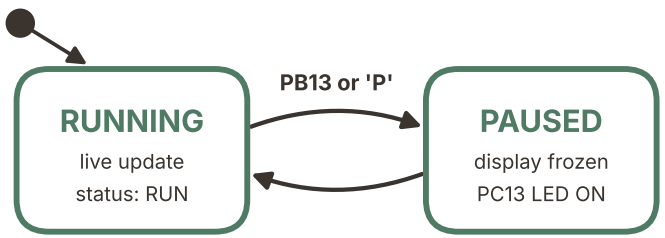
## ACTIVE - six regions run side by side (any combination is possible)

### 1 DISPLAY MODE



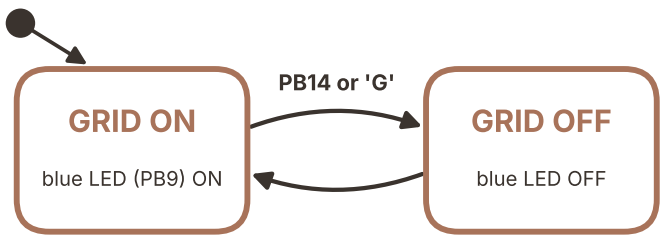

*button press or letter sent from the phone toggles the state*

### 2 RUN / PAUSE



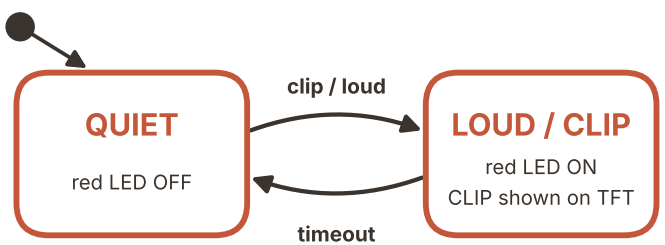

*button press or letter sent from the phone toggles the state*

### 3 GRID



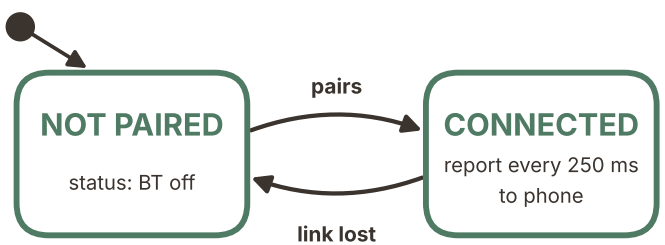

*button press or letter sent from the phone toggles the state*

### 4 LEVEL (red LED PB7)




*ADC hits 0 or 4095 = loud; red LED stays on for a short hold time*

### 5 BLUETOOTH LINK (PB15)




*HC-05 STATE pin (PB15) high = connected*

### 6 GAIN (+ / - from phone)

*shown on TFT status line: GAIN +x dB*

7 gain levels, 6 dB per step
stops at both ends (-12 and +24)




Heartbeat: green LED (PB8) blinks all the time in ACTIVE, so you can see the firmware is alive. Start-up values: BARS, RUNNING, GRID ON.